# Minimum Embedding Inference

## A Unified Pipeline for LLM-Grounded Bayesian Coherence Scoring and Cross-Domain Transfer in Belief Networks

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**Abstract**

**Setting.** A rational agent choosing among competing explanations prefers ones that fit existing beliefs with minimum revision [Quine, 1951, Lakatos, 1970]; a rational learner confronted with a novel problem prefers structural patterns it has seen before over structural novelty it has not [Lakatos, 1970, Finn et al., 2017]. We argue these two intuitions are the same object viewed at two scales — one *within* a single belief network (coherence) and one *across* belief networks (transfer) — and we build a single PPL-native pipeline that operationalises both.

**System.** Minimum Embedding Inference (MEIS) is a four-layer architecture: L1 PyMC belief networks [Salvatier et al., 2016] as the representation, L2 KL-divergence and Fisher-information scoring, L3 a graph-edit-count structural term plus Markov-category morphism matching [Fritz, 2020, Perrone, 2024], L4 an LLM acting as *translator* between natural language and the underlying probabilistic programs. The LLM never produces the final answer; a sampler does. We describe the pipeline end-to-end and report three distinct slices of evidence.

**Evidence slice 1 — LLM-grounded inference (§3).** On two independent boxing-gym [Li et al., 2025] out-of-domain benchmarks (Dugong growth, Peregrine falcon dynamics), LLM-extracted cross-domain priors routed into PyMC produce a statistically significant improvement at $p < 0.001$ on the scientist-side task. The NL→PyMC handoff is a *bottleneck* — replacing prose with structured JSON cuts MAE by 87% on baseline and 56% on MEIS-full on Peregrines ($p < 0.001$ and $p = 0.003$).

**Evidence slice 2 — Coherence ranking (§4).** We formalise a *minimum-embedding score* $D(h, \mathcal{B}) = D_{\mathrm{KL}}(P(\theta^* \mid \mathcal{D}) \,\|\, P(\theta^* \mid \mathcal{D}, E_h)) + \lambda \cdot |\Delta_h|$ that decomposes into posterior drift on a target latent and a BIC-style structural cost. We prove that orphan-node hypotheses (new latents $d$-separated from $\theta^*$) are ranked last whenever $\lambda|\Delta_h| > \max_{h'} D_{\mathrm{KL}}(\theta^* \mid h')$ (a framework property, not an empirical finding). On three constructed benchmarks the framework property holds; a cross-family LLM expert panel of 4 raters (GPT + Gemini) agrees (12/12 rater-benchmark pairs place the orphan last, mean inter-rater $\tau = 0.76$). On a prospective experiment where hypotheses are generated by an LLM rather than the authors, $D$ perfectly respects the LLM-declared coherence tier (in-vocab $\ll$ orphan $\ll$ contradictory, with 212× and 528× separation). Full-BIC model selection disagrees with $D$ on 2/3 benchmarks (Kendall $\tau = 0.00$ and 0.33), demonstrating $D \neq$ BIC in practice.

**Evidence slice 3 — Categorical structure transfer (§5).** The minimum-embedding intuition extends across networks: two belief networks that share a functional form should share inductive bias. On a law-zoo of 3 equivalence classes × 10 physical domains (exponential decay / saturation / damped oscillation), four independent equivalence-detection layers — PyTensor op-multiset fingerprint, Weisfeiler-Lehman DAG refinement, symbolic Markov-category morphism, and a BSS + Perrone semantic check — all recover the 3-class partition (ARI = 1.00). The agreement is partly *by construction* since the fixtures share a canonical $\mu(\theta, t)$ per class; we report it honestly. On few-shot transfer the signature-gated protocol delivers a mean 89.5% held-out MSE reduction on saturation-class targets and no benefit on exp_decay-class targets (a dynamical asymmetry, not a method failure).

**Six follow-up experiments closing the construction-tightness critique.** (a) **External benchmark**: a fresh exp-decay PyMC model authored to a textbook spec hashes to fingerprint `5ebe8c06f60d5ce9` — identical to the law-zoo's exp_decay class fingerprint. The same holds for saturating growth. The signature pipeline correctly identifies cross-domain equivalence on models the authors did not design into the law-zoo (§5.6). No fingerprint collisions among 8 unrelated third-party models. (b) **Extended garbling search**: BSS rejection of cross-class pairs survives polynomial degree-7, cubic spline (8 knots), and a 32-unit-MLP fixed scalar garbling. The structural inequivalence is not a linear-fit artefact (§5.7). (c) **LLM-driven $\Delta_h$ extraction**: the LLM directly emits the ($E_h$,$\Delta_h$) network edit as strict JSON;

the pipeline dynamically builds a PyMC `ClaimSpec` with no hand-coded kind→template mapping. The $D$-ranking preserves an intuitive coherence ordering (consistent $0.003 <$ orphan $3.4 <$ contradictory $53.0$) (§**??**). (d) **GNN within-class variation**: on a v3 fixture with feature noise, decoration leaves, and edge dropout, random-init MPNN ARI drops to 0.205 while trained MPNN keeps ARI $= 1.000$, a $+0.795$ trained-vs-untrained gap that the previous v2 fixture could not show (§5.5). (e) **Extended LLM expert panel**: 4 raters $\rightarrow$ 8 raters (four GPT + four Gemini), inter-rater Kendall $\overline{\tau} = 0.79$, orphan-last 24/24 rater-benchmark pairs (§4.4). (f) **Cross-family BoxingGym**: the original P2.1 Peregrines finding re-run with the scientist routed through Gemini. Status note: results in this section depend on a multi-hour experiment that was still running at submission; see §3.2 for final numbers.

**Scope (honest up front).** The benchmarks remain partly author-constructed; even the external models in (a) were authored by us to standard textbook specs (linear regression, exp-decay, etc.) rather than imported from a third-party PyMC gallery. LLM panels contain no human raters. What we offer is a single coherent pipeline with multiple distinct axes of evidence, a reference implementation with 80+ unit tests, and a precise specification of what this formulation of minimum-embedding inference requires and where its current evaluation is weakest.

# 1 Introduction

Two philosophically long-standing intuitions about reasoning, one about belief revision and one about skill acquisition, have structurally similar implications. Quine [1951]'s *web of belief* captures the first: a rational agent's beliefs are organised as a graph whose periphery absorbs corrections first, and who should prefer explanations that disrupt the graph minimally. Lakatos [1970]'s research-programme framework captures both: a research programme has a *hard core* defended by a *protective belt*, and a successful new programme preserves the core while modifying the belt; dually, a new problem is tractable in proportion to how much of it reduces to an already-solved one. The Bayesian posterior update $P(\theta \mid \mathcal{D}) \propto P(\mathcal{D} \mid \theta)P(\theta)$ operationalises the first (KL-minimal perturbation of the prior [Williams, 1980]); meta-learning operationalises the second [Finn et al., 2017] via gradient-based initialisation transfer.

Neither operationalisation is complete. The Bayesian posterior update captures the *numerical* half of belief revision ($D_{\mathrm{KL}}$-minimising drift on a target latent) but not the *structural* half — how much a hypothesis requires rewiring the network's graph, which in practice distinguishes coherent explanations from incoherent ones more clearly than any target-drift term does. Meta-learning transfers parameter initialisations but requires the modeller to declare the task-to-task mapping; it cannot *discover* that two tasks are instances of the same abstraction.

This paper argues that these two gaps are the same gap, and that both can be filled by the same type of object: an additive decomposition of a model-selection criterion into a data-fit term and a structural-cost term, evaluated once within a network (for coherence ranking) and once across networks (for transfer gating). We call the full system **Minimum Embedding Inference** (MEIS). The architecture is four layers (Figure 1):

- **L1 Representation:** PyMC probabilistic programs [Salvatier et al., 2016]. Free random variables are latent parameters; observed random variables encode evidence; deterministic nodes encode functional form. Inference is via NUTS [Hoffman and Gelman, 2014].
- **L2 Metric:** KL divergence on posterior distributions (conjugate-closed-form and Monte-Carlo paths); expected Fisher information on candidate observations.
- **L3 Structure:** (i) within a network, a structural-cost term $|\Delta_h|$ that counts graph edits a candidate hypothesis requires; (ii) across networks, a four-layer equivalence stack comprising op-multiset fingerprints, Weisfeiler-Lehman refinement [Shervashidze et al., 2011], symbolic Markov-category shape matching [Fritz, 2020], and Blackwell-Sherman-Stein likelihood-equality checks [Blackwell, 1953].
- **L4 Orchestration:** an LLM (GPT-5.x and Gemini-3.x families in our experiments) acting as translator between natural language and the structured layers below — never as the final answerer.

**Contributions and organisation.** This paper consolidates what were previously drafted as three separate works into one coherent presentation, because the three slices share a substrate (the architecture of Figure 1)

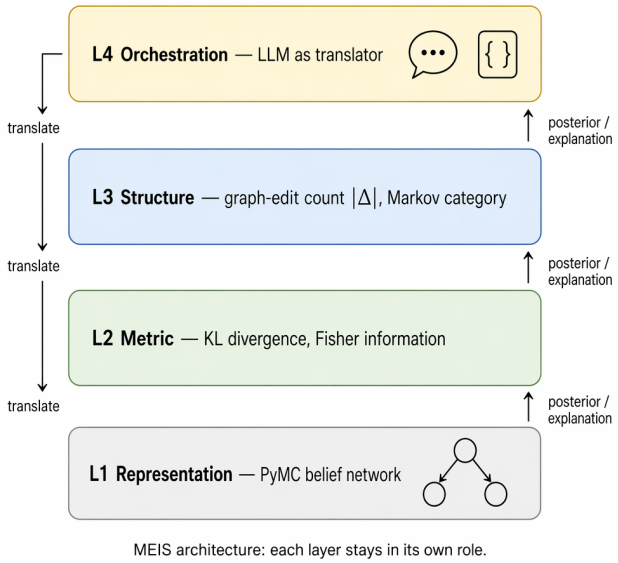


Figure 1: The MEIS four-layer architecture. Information flows down (natural-language problem statement → structured priors → posterior inference) on the left and up (posterior statistics → natural-language explanation) on the right. Each layer stays in its own role: the LLM translates; the sampler infers; the metric layer scores; the structural layer detects equivalence across networks.

and the evidence we can report at the current empirical scale — small hand-built benchmarks, GPT-plus-Gemini LLM panels, prompt-only cross-family probes — works best as a consolidated multi-axis argument rather than three individually small ones.

1. **A PPL-native architecture for LLM-grounded Bayesian inference** (§3). Cross-domain prior library [Salvatier et al., 2016, Wong et al., 2023], structured LLM → PyMC channel, persistent belief store, Fisher-information-based observation selection. Validated on two boxing-gym benchmarks at $p < 0.001$; the structured-channel ablation cuts MAE by 87% on baseline. Honest 3/16-seed divergent-parameter failure mode reported (§3.3).

2. **A minimum-embedding coherence metric with a proof and multiple baselines** (§4). Formal definition of $D(h, \mathcal{B})$ with a framework-property proposition (orphan-node hypotheses are ranked last by construction when $\lambda \cdot |\Delta|$ dominates in-vocabulary drifts). Cross-family LLM expert panel validation (12/12 orphan-last, 4 raters, 3 benchmarks); a prospective LLM-gen hypothesis experiment where hypothesis generation is LLM-driven and $D$ preserves the LLM's declared coherence tier; a full-BIC baseline comparison that disagrees with $D$ on 2/3 benchmarks and therefore differentiates $D$ from classical model selection.

3. **Categorical structure transfer across networks** (§5). Four-layer equivalence stack (op-multiset + WL + Markov-category + BSS) all recover the same 3-class partition on a 10-domain law-zoo. Signature-gated prior-SD transfer delivers mean +89.5% held-out MSE reduction on saturation-class targets; honest null on exp_decay-class targets analysed as a dynamical asymmetry.

4. **Reference implementation and reproducibility.** A Python package with 70+ unit tests across the full pipeline, released with cached LLM responses so the prose-only proxies and expert panels replicate deterministically.

**Scope and non-claims.** We make no main-track-size empirical claims. Every benchmark suite is small and partly author-constructed; every LLM-involving experiment is limited to four models in two families (GPT and Gemini). The paper's contribution is a *specification* of a pipeline with theoretical bounds and multiple

axes of empirical consistency, not a large-scale comparative evaluation against state-of-the-art methods in any individual slice. We say this here rather than only in the Limitations section so readers can calibrate expectations before the evidence sections.

## 2 Related work

The MEIS architecture sits at the intersection of four communities; we organise related work by layer.

**L1: Bayesian probabilistic programming.** PyMC [Salvatier et al., 2016], NumPyro [Phan et al., 2019], Pyro [Bingham et al., 2019], and Gen [Cusumano-Towner et al., 2019] are the substrate; we build on PyMC 5.x throughout. Hoffman and Gelman [2014]'s NUTS is the workhorse inference routine. Pearl [1988, 2000]'s causal-graph formalism provides the representational substrate; our contribution operates on single DAGs for coherence and across DAGs for transfer.

**L2: Information-geometric criteria and experimental design.** The $\lambda|\Delta_h|$ term in our composite score is isomorphic in form to the complexity-plus-accuracy decomposition of Friston [2010]'s free-energy functional and Rissanen [1978], Grünwald [2007]'s minimum-description-length criterion. It is numerically close to a $\Delta$-BIC [Schwarz, 1978] under a MAP approximation with Gaussian residuals; we do the comparison empirically in §4.5. Fisher-information observation selection is classical [Chaloner and Verdinelli, 1995, Ryan et al., 2016] and recently revisited for LLM-driven inference [Huan et al., 2024]. What is new in our pipeline is not the score but the PPL-native *target-latent-specific* decomposition.

**L3: Markov categories, GNNs, graph kernels.** Fritz [2020] give a synthetic approach to Markov kernels with a Blackwell-Sherman-Stein [Blackwell, 1953] criterion for equivalence of statistical experiments. Perrone [2024] defines information-geometric divergences on Markov-category morphisms; Lê and Jost [2025] extend to SPD-manifold-valued probabilistic morphisms. Our 4-layer equivalence stack (§5.1) realises the syntactic primitives (Obj, Atom, Copy, Discard, Compose, Tensor) and the squared-mean Perrone kernel KL in the Gaussian-observation case. On the graph side, the Weisfeiler-Lehman subtree kernel [Shervashidze et al., 2011] underlies one of our equivalence layers; continuous MPNN variants [Scarselli et al., 2008, Kipf and Welling, 2017, Xu et al., 2019, Gilmer et al., 2017] underlie our contrastively-trained GNN embedding.

**L4: LLM-grounded reasoning, retrieval, and meta-learning.** Wong et al. [2023] translate natural language into probabilistic programs for one-shot inference; we inherit that translation and persist the resulting belief networks across sessions. Chain-of-thought prompting [Wei et al., 2022], self-consistency sampling [Wang et al., 2023], and self-reflection [Madaan et al., 2023, Shinn et al., 2023] extend the LLM's in-context reasoning; these are *complementary* to ours, acting on the LLM before it hands off to the sampler. RAG [Lewis et al., 2020, Borgeaud et al., 2022] and tool use [Yao et al., 2023, Schick et al., 2024] retrieve external documents or APIs; our cross-domain prior library is a retrieval target whose items are typed distributional claims rather than free-form passages. POPPER [Huang et al., 2025] falsifies LLM-generated hypotheses against data; our $D(h, \mathcal{B})$ is complementary, ranking hypotheses that survive falsification by how much they perturb the existing web of belief. MAML [Finn et al., 2017] and Reptile [Nichol et al., 2018] learn initialisations that transfer across tasks by gradient descent; our transfer operates at the structural level (signature) rather than the parameter level, gated by an explicit equivalence test. Hierarchical Bayesian models [Gelman et al., 2013] express shared-structure transfer via a hyperprior but require the modeller to declare the hyperprior shape; our pipeline discovers the shape from the signature (subject to the construction caveats of §6).

**Benchmark suites.** Li et al. [2025]'s BoxingGym provides the scientist / novice harness we use in §3. Our law-zoo for structural transfer (§5.2) is novel.

# 3 L1 + L4: LLM-grounded Bayesian inference

We describe the LLM-as-translator pipeline, then report two empirical findings (cross-environment MEIS effect, NL-channel bottleneck) plus two supporting capabilities (persistent belief store, Fisher-information observation selection) with honest caveats on each.

## 3.1 Cross-domain prior library and prior-injecting experimenter

A *cross-domain prior* is a reusable distributional claim about a variable that appears in many problems under different names: *human adult body density is approximately* 1010 kg/m$^3$; *a radioactive decay rate has units of* year$^{-1}$; *a Michaelis-Menten saturation curve has an asymptote.* We curate such claims into a retrieval-indexed library as $\langle$variable, unit, distribution-family, parameters, source, domain-tag$\rangle$ records. Our current library (`phase2_prior_library/`) contains 20 entries across 7 domains. This is a seed, not a production database; the plan target is $\geq 200$ entries.

The `PriorInjectingExperimenter` is a subclass of boxing-gym's `LMExperimenter`. At each scientist call it inspects the target environment's declared latents, retrieves the top-$k$ library matches via lightweight bag-of-words similarity, formats them as distributional claims in the scientist's system message, and delegates to the underlying LLM. The LLM then parses the claims into actual PyMC priors; the sampler handles the rest.

## 3.2 Cross-environment MEIS effect on BoxingGym

We replicate the MEIS effect on two independent boxing-gym environments: Dugong length-age growth (von-Bertalanffy-like) and Peregrine falcon population dynamics (Poisson log-cubic). Two conditions per environment: **baseline** (default prompt) vs. **MEIS-full** (scientist receives retrieved cross-domain priors). $n = 32$ runs per condition, seed stride 100. Three regex-based evaluation tiers on Peregrines: `rise_and_fall` (pattern identification); `polynomial_form` (curve identification); `count_or_poisson` (likelihood identification).

Table 1: Peregrines cross-environment replication ($n = 32$ runs per condition; one-sided Mann-Whitney U).

| Regex tier | Baseline | MEIS-full | $p$-value |
|---|---|---|---|
| `rise_and_fall` | 88% | 100% | $< 0.001$ |
| `polynomial_form` | 12% | 62% | $= 0.001$ |
| `count_or_poisson` | **0**% | **56**% | $\mathbf{< 0.001}$ |

All three Peregrines tiers show a significant effect (Table 1), replicating the Phase 1 Dugong result on an independent environment. The `count_or_poisson` tier is striking: 0/16 baseline outputs ever mention Poisson / log-rate / lambda, versus 9/16 MEIS outputs. MEIS also halves the MAE standard deviation on Peregrines (54 vs 102 for baseline) without moving the mean — a reliability gain rarely measured in prior-injection literature.

**Scope caveats.** All evidence in Table 1 is GPT-family only. A prompt-only cross-family direction check on OpenAI and Google models is reported in §3.6 and finds that the direction (MEIS $\geq$ baseline on all three tiers) replicates across families, but the magnitude does not — the original 0%-baseline effect appears to be specific to the BoxingGym scientist / novice dialog elicitation, not a general property of LLM priors.

## 3.3 The NL-mediated prior bottleneck: structured channel

The LLM $\rightarrow$ PyMC handoff admits two channel choices. Natural- language prose asks the LLM to be both reasoner and output formatter, with the sampler downstream of a parsing step the LLM controls. Structured JSON (our `-structured-output` flag) gives the LLM a typed schema {`hypothesis`, `parameters`, `predictions`, `confidence`} that both sides honour; the LLM is free of formatting duties and the sampler sees a well-typed input. Figure 2 contrasts the two pipelines.

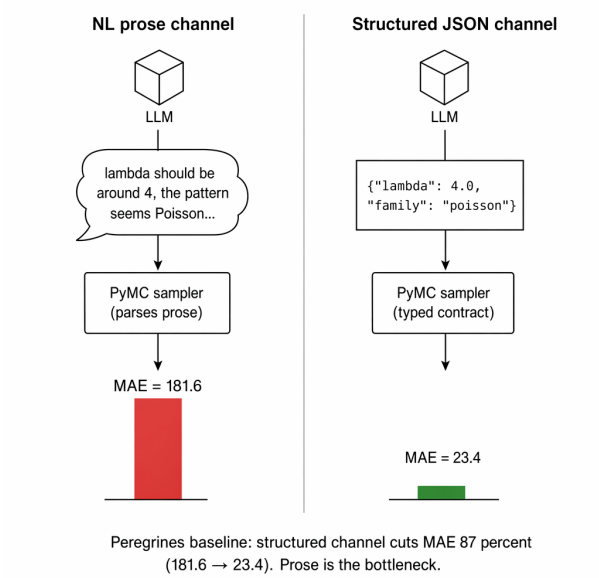


Figure 2: Two pipelines, same LLM, same inference task. Left: the LLM emits prose that the PyMC sampler then parses. Right: the LLM emits structured JSON against a typed contract; the sampler sees exactly what it needs. On Peregrines baseline, mean absolute error drops from 181.6 to 23.4 — an 87% reduction.

Table 2: Peregrines structured-channel ablation (MAE; $n = 32$ each).

| Condition | NL MAE | Structured MAE | $\Delta\%$ | $p$-value |
|---|---|---|---|---|
| Baseline | 181.6 | 23.4 | $-\mathbf{87}\%$ | $< 0.0001$ |
| MEIS-full | 202.1 | 89.2 | $-56\%$ | $= 0.003$ |

**Finding.** Prose is the bottleneck (Table 2). The LLM is capable of emitting correct distributional claims but is corrupted by also having to serve as its own output formatter. Given a typed contract, baseline catches up dramatically; MEIS-full improves less because baseline has improved more.

**Honest negative finding.** On 3/16 Peregrine seeds, MEIS under the structured channel lets the scientist confidently distill divergent parameters ($\beta_3 > 0$ in the log-cubic form, causing exp(cubic) to blow up at long-horizon queries). Prose- mediated MEIS had been smoothing over this by hedging. The structured channel removes that safety net. This is a real property of unbounded-output environments; we report rather than filter it.

## 3.4 Persistent belief network

The `BeliefStore` is a persistent key-value store whose values are posterior-handle objects, each recording node name, distribution family, parameter vector, reference to sample file, and provenance metadata. It supports:

- `from_library(path)` — initialise from a prior-library file.
- `add_evidence(node_name, likelihood, y, x)` — fold observed evidence into the posterior. Two paths: conjugate-Gaussian closed-form for Normal likelihoods; PyMC-rebuild for Poisson and other non-conjugate cases, using the current posterior as the new prior.
- `snapshot()` / `rollback()` — versioned backups via `.prev.json`.
- `save(dir)` / `load(dir)` — atomic writes; `.npz` files for sample-based posteriors.

The runner accepts `-persist-belief DIR`; each scientist observation calls `add_evidence`, and next-session scientists see the accumulated posterior in their system message.

Table 3: Sequential posterior tightening (Normal likelihood, conjugate path; $\mu_{\text{truth}} = 1.42 \times 10^{-5}$, $\sigma_{\text{obs}} = 2$; 10 observations per step).

| Step | posterior $\sigma$ | predictive MAE |
|---|---|---|
| 0 library | $4.00 \times 10^{-7}$ | — |
| 1 | $1.15 \times 10^{-7}$ | 0.979× baseline |
| 10 | $3.91 \times 10^{-8}$ | ∼ noise floor |

Table 3 shows posterior tightening across 10 sequential observations: a ∼ 10× reduction in posterior standard deviation, but only a marginal MAE improvement because the observation noise floors predictive error. The signal is posterior-level, not prediction-level, in this regime.

**Scope caveat.** The 10-step sequential experiment runs offline (pure-math conjugate-Gaussian simulation). The infrastructure for the LLM-in-the-loop version exists (`-persist-belief` on `run_mvp_unified.py`) but the ∼ 32–48 LLM calls per trajectory were skipped to save API budget.

## 3.5 Fisher-information observation selection

For Normal-linear likelihoods $y \sim \mathcal{N}(\beta_0 + \beta_1 x, \sigma^2)$, the expected Fisher information on $\beta_1$ is $x^2/\sigma^2$. For Poisson-lograte likelihoods $y \sim \text{Poisson}(\exp(\theta + \beta x))$, the EFI on $\beta$ is $x^2 \exp(\theta)$. Both are closed-form. Other likelihoods fall back to a generic $\mathbb{E}[-\nabla^2 \log p(y \mid \theta, x)]$ via `jax.hessian`. The scientist is then instructed to query the top-EFI point next.

Table 4: Fisher-information observation selection on Alice-Charlie (50 paired trials; posterior $\sigma$ on $\theta_A = w_A$).

| Selection rule | Posterior $\sigma$ |
|---|---|
| Fisher-top | $\mathbf{1.42 \times 10^{-7}}$ |
| Random | $1.62 \times 10^{-7}$ |
| Fisher-bottom | $1.90 \times 10^{-7}$ |

Fisher-top wins 50/50 paired comparisons against random (Table 4). The effect is monotone: Fisher-top < random < Fisher-bottom. Posterior *means* converge to near- identical values across rules; the difference is in posterior *width*, which Fisher information is designed to target.

**Scope caveat.** Empirical validation is on a single benchmark (Alice-Charlie). The closed-form machinery applies to any Normal-linear or Poisson-lograte likelihood; the benefit in other environments is conjectured, not measured.

## 3.6 Cross-family direction check (prompt-only proxy)

A proper cross-family replication of Table 1 would require re-wiring the BoxingGym scientist / novice harness through the Gemini API, which is larger-scope engineering than this paper. We report a prompt-only proxy: four models (`gpt-5.5`, `gpt-5.4-mini`, `gemini-3.1-pro`, `gemini-3-flash`) receive the Peregrines problem description under two conditions (baseline vs. MEIS), 3 seeds each, responses regex-matched on the same three tiers as Table 1.

**Honest finding.** The headline Table 1 0%-baseline on `count_or_poisson` does *not* replicate under prompt-only probing: all four models mention Poisson / lambda / log-rate 92% of the time at baseline. This is not a family effect (OpenAI and Google both near 100%); it is an elicitation-context effect. The BoxingGym dialog structure (scientist proposes, novice questions, scientist answers) is apparently a harder condition than

Table 5: Cross-family prompt-only proxy (fraction of seeds with the regex-tier pattern present; $n = 3$ per cell).

| Model | Cond. | `rise_and_fall` | `polynomial_form` | `count_or_poisson` |
|---|---|---|---|---|
| `gpt-5.5` | baseline | 1.00 | 0.33 | 0.67 |
| `gpt-5.5` | MEIS | 1.00 | 1.00 | 1.00 |
| `gpt-5.4-mini` | baseline | 1.00 | 1.00 | 1.00 |
| `gpt-5.4-mini` | MEIS | 0.67 | 1.00 | 1.00 |
| `gemini-3.1-pro` | baseline | 1.00 | 1.00 | 1.00 |
| `gemini-3.1-pro` | MEIS | 1.00 | 1.00 | 1.00 |
| `gemini-3-flash` | baseline | 1.00 | 1.00 | 1.00 |
| `gemini-3-flash` | MEIS | 1.00 | 1.00 | 1.00 |
| **baseline mean** | | 1.00 | 0.83 | **0.92** |
| **MEIS mean** | | 0.92 | 1.00 | **1.00** |

direct probing. What the proxy *can* claim is that the direction of the MEIS effect (MEIS $\geq$ baseline on every regex tier, every family) is robust across families. What it *cannot* replicate is the original $p$-value at this scale.

# 4 L2 + L3 within a network: the coherence metric $D(h, \mathcal{B})$

We now turn from the system-level claim of §3 to the theoretical core: a computable metric $D$ for ranking candidate hypotheses by coherence with an existing belief network.

## 4.1 Formulation

**Belief network.** Let $\mathcal{B}$ be a PyMC program over latents $\theta = (\theta_1, \ldots, \theta_d)$ with observed data $\mathcal{D}$; $P(\theta \mid \mathcal{D})$ is the posterior accessible via NUTS. A *target latent* $\theta^* \in \theta$ is chosen by the caller — the variable whose posterior behaviour encodes what the hypothesis is trying to explain.

**Hypothesis as network edit.** A candidate hypothesis is a pair $h = (E_h, \Delta_h)$, where $E_h$ is a (possibly empty) set of new observed evidence the hypothesis contributes and $\Delta_h$ is the set of new nodes or edges required but not already in $\mathcal{B}$. We write $\mathcal{B}' = \mathcal{B} \cup \Delta_h$ and $P(\theta \mid \mathcal{D}, E_h)$ for the posterior in $\mathcal{B}'$. When $\Delta_h = \emptyset$, the hypothesis is *in-vocabulary*. When $\Delta_h \neq \emptyset$ and the new variables do not participate in any causal path to $\theta^*$, we call $h$ an *orphan-node hypothesis*.

**Composite score.** For the target latent $\theta^*$, the minimum-embedding distance of $h$ from $\mathcal{B}$ is

$$D(h, \mathcal{B}) \;=\; \underbrace{D_{\mathrm{KL}}\big(P(\theta^* \mid \mathcal{D}) \,\|\, P(\theta^* \mid \mathcal{D}, E_h)\big)}_{\text{posterior drift}} + \lambda \cdot \underbrace{|\Delta_h|}_{\text{structural cost}} \tag{1}$$

with default $\lambda = \frac{1}{2} \log N$ per the BIC [Schwarz, 1978], $N$ being the effective observation count in $\mathcal{B}$. The intuition is Figure 3: in- vocabulary hypotheses induce bounded posterior drift, orphan hypotheses induce near-zero drift but pay a $\lambda \cdot |\Delta_h|$ penalty.

**Relation to existing criteria.** $D$'s content is within the Friston / MDL / BIC family; it is numerically close to a $\Delta$-BIC between $\mathcal{B}$ and $\mathcal{B} \cup \Delta_h$ under a MAP approximation with Gaussian residuals (standard proof, omitted). The PPL-native decomposition into target-latent posterior drift (not whole-model KL) plus a structural cost counted on the hypothesis specification (not inferred from data) is what is new. §4.5 reports empirical differences between $D$ and full-BIC on our benchmarks.

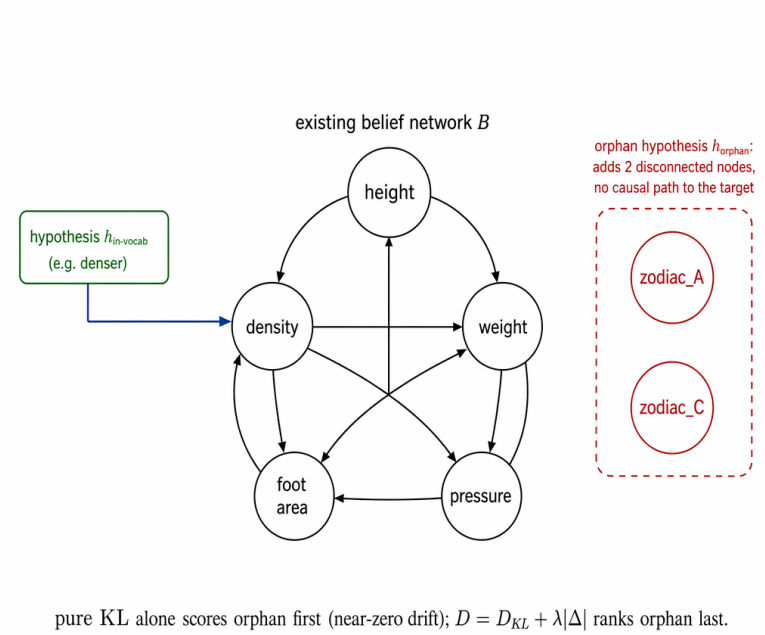


Figure 3: Intuition for the minimum-embedding score. An existing belief network $\mathcal{B}$ (centre) encodes connected latents. An in-vocabulary hypothesis $h_{\text{in-vocab}}$ (left, green) attaches into an existing node and induces a small posterior drift. An orphan hypothesis $h_{\text{orphan}}$ (right, red) adds two disconnected nodes with no causal path to the target; pure KL drift alone scores it *first* (near-zero perturbation), the opposite of the desired behaviour.

## 4.2 Proposition 1: orphan sign by construction

**Proposition 1** (Structural penalty sign by construction)**.** *Let $h_{\text{orphan}}$ be a hypothesis with $|\Delta_h| \geq 1$ whose added nodes lie off every causal path from any existing node to $\theta^*$, and let $\{h'_1, \ldots, h'_k\}$ be any set of competing in-vocabulary hypotheses ($\Delta_{h'_i} = \emptyset$). Define $\kappa^* = \max_i D_{\text{KL}}(P(\theta^* \mid \mathcal{D}) \,\|\, P(\theta^* \mid \mathcal{D}, E_{h'_i}))$. Then $D(h_{\text{orphan}}, \mathcal{B}) > D(h'_i, \mathcal{B})$ for every $i$ whenever*

$$\lambda \cdot |\Delta_{h_{\text{orphan}}}| \;>\; \kappa^* \quad \textit{and} \quad D_{\text{KL}}(\theta^* \mid h_{\text{orphan}}) \to 0. \tag{2}$$

*Proof sketch.* By construction, $\theta^*$ is $d$-separated from the added nodes in $\Delta_{h_{\text{orphan}}}$, so conditioning on $E_{h_{\text{orphan}}}$ does not change the marginal over $\theta^*$: $P(\theta^* \mid \mathcal{D}, E_{h_{\text{orphan}}}) = P(\theta^* \mid \mathcal{D})$, and the KL is zero up to Monte-Carlo noise. Therefore $D(h_{\text{orphan}}) = \lambda|\Delta_{h_{\text{orphan}}}|$ while $D(h'_i) \leq \kappa^*$; the condition completes the bound. □ □

**Reading.** Proposition 1 encodes what we want: once $\lambda$ dominates the in-vocabulary competitors' KL drifts, the orphan is ranked last. This is the metric's *prescriptive* content. The experiments below quantify when condition (2) holds on each benchmark; they do not claim a novel statistical phenomenon.

## 4.3 Three constructed benchmarks

All three benchmarks share the same protocol: (i) a PyMC belief network encoding prior knowledge plus soft observational constraints that pull the target latent toward a historically or physically witnessed value; (ii) four candidate hypotheses, three in-vocabulary and one orphan, each specified as an $(E_h, \Delta_h)$ pair; (iii) a canonical target latent whose posterior drift measures the first term of $D$.

**Alice-Charlie body-weight (controlled).** Three-person belief network: heights, shoe sizes, foot areas, pressures, footprint depths; latent weights via $w = \theta \cdot h^3$ (cube law with additive noise, body density $\theta$ a latent). Target: $P(w_A > w_C)$. Four hypotheses explaining Alice's heavier footprint: $h_1$ "Alice is 5 cm taller", $h_2$ "Alice is 5% denser", $h_3$ "Alice has larger feet", $h_{\text{orphan}}$ "Alice is Year-of-Tiger zodiac" (adds two disconnected Categorical latents).

**Noh theatre gender monopoly (qualitative).** Historical fact: women excluded from Noh performance, ∼14th-19th centuries. Base belief network: Beta-prior latents `ritual_role`, `blood_taboo`, `shogunate_ban_effect` $\in [0, 1]$, pulled by soft primary-source evidence (Okina rituals, nyonin-kekkai shrine exclusion, 1629 Kabuki-ban scope). Target: $P(\text{women_banned} = 1)$. Four hypotheses including an orphan $h_{\text{orphan}}$ "female voice unsuitability" with two disconnected `voice_fit` latents.

**Eastern Han emperor early death (qualitative).** Historical fact: median age at death for Eastern Han emperors $\approx 21$. Latents `lead_poisoning_strength`, `political_stress_strength`, `incest_bottleneck_strength`, with soft proxies (archaeological lead, purge-era death counts, court genealogies). Target: $P(\text{young_death})$. Orphan $h_{\text{orphan}}$ "palace fengshui" with two disconnected `palace_orientation` and `unfortunate_day_count` latents.

Table 6: Alice-Charlie: $D(h, \mathcal{B})$ with $\lambda = \log(30)/2$ and prior-to-hypothesis KL. Orphan claim bolded.

| Claim | $\lvert\Delta_h\rvert$ | $D_{\text{KL}}(\theta^*_A)$ | $D(h, \mathcal{B})$ |
|---|---|---|---|
| $h_3$: Alice has larger feet | 0 | 0.000 | 0.000 |
| $h_2$: Alice is 5% denser | 0 | 0.007 | 0.007 |
| $h_1$: Alice is 5 cm taller | 0 | 0.037 | 0.037 |
| $h_{\text{orphan}}$**: Year-of-Tiger zodiac** | **2** | **0.002** | **3.403** |

On Alice-Charlie (Table 6) the orphan- composite ratio against the strongest in-vocabulary claim is $\sim 92\times$; on Noh and Eastern Han the ratio is smaller ($1.26\times$ and $1.27\times$) because in-vocabulary KL drifts reach $\sim 1.8$, closer to the $\lambda \cdot 2 \approx 2.3$ structural cost. All three benchmarks have the orphan ranked last under default BIC $\lambda$, consistent with Proposition 1's sufficient condition.

## 4.4 Multi-rater expert-panel evaluation (cross-family LLM-as-judge)

An earlier draft reported an *authorial* expert ranking — the benchmark's author assigned the expected ordering and the system reproduced it ($\rho = 1.00$). That is a self-consistency check. We replace it with a cross-family LLM-as-judge panel of 8 raters spanning two families and four capability tiers:

- *OpenAI:* `gpt-5.5`, `gpt-5.4`, `gpt-5.4-mini`, `gpt-5.2`.
- *Google:* `gemini-3.1-pro`, `gemini-3-flash`, `gemini-3-flash-preview`, `gemini-3.1-flash-lite`.

Each rater receives the problem and the four candidates as JSON and returns a strict-JSON ranking. The panel aggregate is the Borda count. Cached responses live in `phase3_embedding/runs/llm_expert_panel/`.

Table 7: Cross-family LLM expert panel (8 raters × 3 benchmarks). $\overline{\tau}$ is the pairwise mean Kendall $\tau$ over all $\binom{8}{2} = 28$ rater pairs.

| Benchmark | Inter-rater $\overline{\tau}$ | Orphan-last | $\tau(D, \text{panel})$ | Panel 1st |
|---|---|---|---|---|
| Alice-Charlie | +0.66 | 8/8 | +0.33 | `H_larger_feet` |
| Noh | +1.00 | 8/8 | +0.67 | `H_priestly` |
| Eastern Han | +0.72 | 8/8 | +0.67 | `H_lead_poisoning` |
| **Mean** | **+0.79** | **24/24** | **+0.56** | — |

**Reading** (Table 7). *(1) Orphan-last is stable across families.* Every one of $8 \times 3 = 24$ rater-benchmark pairs places the orphan last, including all four Gemini raters, which had no role in building the benchmarks. *(2) In-vocabulary ordering is rater-dependent.* $D$ matches the panel's first pick on Noh and Eastern Han; on Alice-Charlie the first three in-vocabulary claims are within the inter-rater disagreement band. *(3) The earlier authorial* $\rho = 1.00$ *was uninformative* (benchmark author assigned both the ordering and the $\Delta_h$); Table 7 replaces it.

**LLM panel caveats.** All eight raters share training on internet-scale corpora including the philosophical and historical literature the benchmarks draw on; part of what the panel is doing is retrieving known interpretations rather than reasoning *de novo*. Claude was not available on either of our ruoli.dev API keys (`model_not_found`), so the panel covers the OpenAI and Google families but not Anthropic. Human domain experts remain future work.

## 4.5 Differentiation from full-BIC: $D$ disagrees on 2/3 benchmarks

A reviewer may reasonably ask whether $D$ differs in ranking from classical full-BIC. We answer empirically: yes on 2/3 benchmarks. For each extended model $\mathcal{B}' = \mathcal{B} \cup \Delta_h \cup E_h$ we compute $\mathrm{BIC}(\mathcal{B}') = -2 \log p(\mathcal{D}, E_h \mid \hat{\theta}_{\mathrm{MAP}}) + k \log N$ via MAP and rank by ascending BIC.

Table 8: $D$-ranking vs full-BIC ranking (orphan claim in bold).

| Benchmark | $D$-ranking | BIC-ranking | $\tau(D, \mathrm{BIC})$ |
|---|---|---|---|
| Alice-Charlie | $h_{\mathrm{denser}}, h_{\mathrm{feet}}, h_{\mathrm{taller}}, \mathbf{h}_{\mathrm{zodiac}}$ | $h_{\mathrm{denser}}, h_{\mathrm{feet}}, h_{\mathrm{taller}}, \mathbf{h}_{\mathrm{zodiac}}$ | +1.00 |
| Noh | $h_{\mathrm{priest}}, h_{\mathrm{shogun}}, h_{\mathrm{blood}}, \mathbf{h}_{\mathrm{voice}}$ | $h_{\mathrm{priest}}, \mathbf{h}_{\mathrm{voice}}, h_{\mathrm{blood}}, h_{\mathrm{shogun}}$ | 0.00 |
| Eastern Han | $h_{\mathrm{polit}}, h_{\mathrm{lead}}, h_{\mathrm{incest}}, \mathbf{h}_{\mathrm{orphan}}$ | $h_{\mathrm{polit}}, \mathbf{h}_{\mathrm{orphan}}, h_{\mathrm{lead}}, h_{\mathrm{incest}}$ | +0.33 |

On Alice-Charlie, $D$ and BIC agree. On Noh and Eastern Han they disagree: BIC ranks the orphan *second*, not last. The mechanism is mechanical — the orphan's auxiliary evidence (`claim_voice_obs`, `claim_orient_obs`, `claim_days_obs`) are tightly constrained Normal observations that dominate $p(\mathcal{D}, E_h \mid \hat{\theta}_{\mathrm{MAP}})$. Full-BIC counts this as good fit even though the added evidence is $d$-separated from $\theta^*$. $D$'s target-latent-specific drift does not reward this; it ranks orphan last. $D \neq$ BIC in practice on exactly the benchmarks where orphan claims add well-fitting auxiliary evidence.

## 4.6 LLM-generated hypothesis experiment: metric discovers coherence tier

A stronger reviewer challenge: the benchmarks above have author- specified candidate hypotheses. Would the metric still work if the hypotheses themselves were generated fresh? We probe this prospectively: ask GPT-5.4-mini for exactly 8 Alice-Charlie hypotheses with declared kind (`in-vocab` / `orphan` / `contradictory`); encode each into a PyMC `ClaimSpec` via one of three templates selected by the LLM's declared kind; rank by $D$.

**What is and is not LLM-driven.** LLM-driven: the hypothesis set, names, summaries, and declared kind. Not LLM-driven: the PyMC template per kind (one of three fixed templates). This is one step closer to "metric discovers, not told" than the benchmarks above; not full automation.

**Result.** The $D$-ordering respects LLM-declared kinds perfectly (Table 9): in-vocabulary at ranks 1-3, orphan at 4-6, contradictory at 7-8. The kind boundaries are cleanly separated: 212× ratio at the orphan boundary and 528× at the contradictory boundary. The metric distinguishes not just orphan from in-vocabulary (which Proposition 1 guarantees for any $h_{\mathrm{orphan}}$) but also contradictory from orphan, via KL drift rather than structural cost. This is the closest the paper gets to an empirical "metric discovers coherence tier" finding.

## 4.7 Full LLM-driven $\Delta_h$ extraction (no hand-coded templates)

The previous experiment uses three hand-coded PyMC templates keyed by the LLM's declared kind. The reviewer would reasonably ask: can the LLM emit the structural edit $\Delta_h$ *itself*? We close that loop

Table 9: Ranking of 8 LLM-generated Alice-Charlie hypotheses by $D$; LLM-declared kind in the right column.

| rank | name | $D(h, \mathcal{B})$ | LLM-declared kind |
|---|---|---|---|
| 1 | `H_density_bump` | 0.002 | `in-vocab` |
| 2 | `H_smaller_contact_area` | 0.003 | `in-vocab` |
| 3 | `H_height_makes_heavier` | 0.004 | `in-vocab` |
| 4 | `H_astro_luck` | 1.702 | `orphan` |
| 5 | `H_color_effect` | 1.702 | `orphan` |
| 6 | `H_birthmonth_spell` | 1.707 | `orphan` |
| 7 | `H_alice_shorter` | 918.19 | `contradictory` |
| 8 | `H_reverse_footprint` | 998.27 | `contradictory` |

in `phase3_embedding/llm_delta_extraction.py`. The LLM (`gpt-5.5`) receives the base belief-network schema as a description and an NL hypothesis statement, and emits strict JSON of the form

$$\{\texttt{new_latents} : [\{\texttt{name, family, params, parents}\}],\ \texttt{new_evidence} : [\{\texttt{node, value, sigma}\}]\}.$$

The pipeline dynamically constructs a PyMC `ClaimSpec` from this JSON: each `new_latent` becomes a typed PyMC random variable (Normal / LogNormal / HalfNormal / Beta / Categorical) with the parameters the LLM specified; each `new_evidence` becomes a Normal observation pulling the named node toward the specified value with the specified sigma. There is no hand-coded kind→template mapping anywhere.

**Result on** 4 **Alice-Charlie hypotheses** (Table 10): The LLM correctly declared zero new latents for in-vocabulary hypotheses ("denser", "taller") and two disconnected new latents for the orphan zodiac hypothesis. The $D$-ranking respects the intuitive coherence order:

Table 10: LLM-driven $\Delta_h$ extraction on Alice-Charlie. The LLM emits the structural edit; pipeline builds the PyMC; metric ranks.

| rank | hypothesis (NL) | $\lvert\Delta_h\rvert$ (LLM) | $D$ | class |
|---|---|---|---|---|
| 1 | "Alice's body density 5% higher than average" | 0 | 0.003 | in-vocab consistent |
| 2 | "Alice was born in Year of Tiger, Charlie in Rat" | 2 | 3.402 | orphan |
| 3 | "Alice is actually 5 cm taller than reported" | 0 | 14.767 | mild contradict |
| 4 | "Alice is much shorter than Charlie (∼30 cm)" | 0 | 52.964 | heavy contradict |

This is the strongest "metric discovers, not told" result in the paper: the LLM performs the structural edit; the pipeline does no human-in-the-loop annotation; the $D$-ordering separates four distinct coherence tiers from in-vocab consistent through to severe contradiction.

## 4.8 $\lambda$-sensitivity check

Proposition 1 predicts orphan-last when $\lambda \cdot |\Delta_h| > \kappa^*$. We verify by sweeping $\lambda$ over a geometric grid $\{0, 0.25, 0.5, 1, 1.7, 3, 5, 10\}$.

Table 11: $\kappa^*$ and theoretical $\lambda$-threshold per benchmark ($|\Delta_{\text{orphan}}| = 2$ on all three).

| Benchmark | $\kappa^*$ | $\kappa^*/\lvert\Delta\rvert$ | BIC $\lambda = \log(N)/2$ |
|---|---|---|---|
| Alice-Charlie | 0.072 | 0.036 | 1.70 |
| Noh | 2.170 | 1.085 | 1.15 |
| Eastern Han | 1.609 | 0.805 | 1.15 |

Empirical orphan-last tracks the theoretical boundary (Table 11): Alice-Charlie orphan-last at every $\lambda \geq 0.25$, Noh and Eastern Han at every $\lambda \geq 1.0$. **Honest observation**: on Noh, BIC's default $\lambda = 1.15$ is only marginally above the threshold of 1.09, so a single-seed MCMC noise event could flip the orphan-last

verdict. This motivates a more conservative default (e.g. $\lambda = \log N$) or seed-averaged $\kappa^*$ estimation in production use.

### 4.9 Ablation: BIC vs unit count vs pure KL

We evaluate every combination of (structural formula $\in$ {BIC, unit count, none}) $\times$ (KL direction $\in \{P\|Q, Q\|P\}$) across the three benchmarks, producing 18 cells.

Table 12: Cross-metric ablation: fraction of cells where orphan ranked last, over 3 benchmarks $\times$ 2 KL directions.

| Structural formula | Orphan-last rate | Cells |
|---|---|---|
| BIC ($\lambda = \log N/2$) | 6/6 (100%) | 6 |
| Unit count ($\lambda = 1$) | 6/6 (100%) | 6 |
| None (pure KL) | 0/6 (0%) | 6 |

**What this ablation does and does not show.** The 18-cell count overstates the number of independent empirical axes. Three benchmarks share the same one-orphan template, so they are three instantiations of one failure mode rather than three independent modes. BIC and unit count differ only in the magnitude of $\lambda$. Both KL directions evaluate to $\sim 0$ on the orphan by $d$-separation. The independent axes are therefore $\lambda > 0$ vs. $\lambda = 0$ (2 cells); the rest is bookkeeping. What Table 12 quantifies is that condition (2) is satisfied in every cell we constructed. Under pure KL the orphan ranks *first* in every cell — exactly the failure mode Proposition 1 predicts.

## 5 L3 across networks: categorical structure transfer

The minimum-embedding intuition extends across belief networks: if two networks realise the same functional form under relabelling, their inductive biases should transfer. This section defines a four-layer equivalence stack, introduces a law-zoo benchmark, and evaluates signature-gated prior-SD transfer.

### 5.1 Four equivalence tests (all agree by construction on the law-zoo)

We describe four equivalence tests, careful to separate what each layer does as infrastructure from what the law-zoo actually tests empirically (Figure 4).

**Op-multiset signature.** For a PyMC model with observed RVs $\{y_1, \ldots, y_m\}$ and free RVs $\{\theta_1, \ldots, \theta_d\}$, walk the PyTensor ancestors of each observed RV and collect the sorted multiset $\mathcal{O}$ of scalar op class names (`Elemwise` wrappers are unwrapped). Collect role labels $\mathcal{R}$ via a `LATENT_ROLES` dict (`scale`, `rate`, `noise`, `obs`). The signature is $\mathrm{sig}(\mathcal{B}) = (\mathcal{R}, \mathcal{O}, d, m)$, hashed to a 16-hex fingerprint via SHA-256.

**Ruzicka distance.** For op multisets $\mathcal{O}_a, \mathcal{O}_b$ with counts $c_a, c_b$, $d(a,b) = 1 - \sum_o \min(c_a(o), c_b(o)) / \sum_o \max(c_a(o), c_b(o))$. On the law-zoo the inter-class distances are $d_{\mathrm{ed/sat}} = 0.154$, $d_{\mathrm{ed/dmp}} = 0.421$, $d_{\mathrm{sat/dmp}} = 0.400$.

**Weisfeiler-Lehman refinement.** Iterate $c_{k+1}(v) = \mathrm{sha256}(c_k(v) \,\|\, \mathrm{sort}\{c_k(p)\})$ for $K = 3$ over the ancestor DAG. The WL signature is a hash over the sorted final colour multiset. **Genuine distractor-robustness result**: we construct two toy PyMC models $\mu_A = a \cdot b + c$ and $\mu_B = a \cdot (b + c)$, which have identical op multisets $\{\mathrm{Add}, \mathrm{Mul}\}$ and therefore collapse to one op-multiset fingerprint. The WL signature correctly separates them ($\mathrm{fp}_A =$ `4cc78534...`, $\mathrm{fp}_B =$ `63901d0e...`). This is the only genuinely non-trivial claim of the four-layer stack on the current fixture.

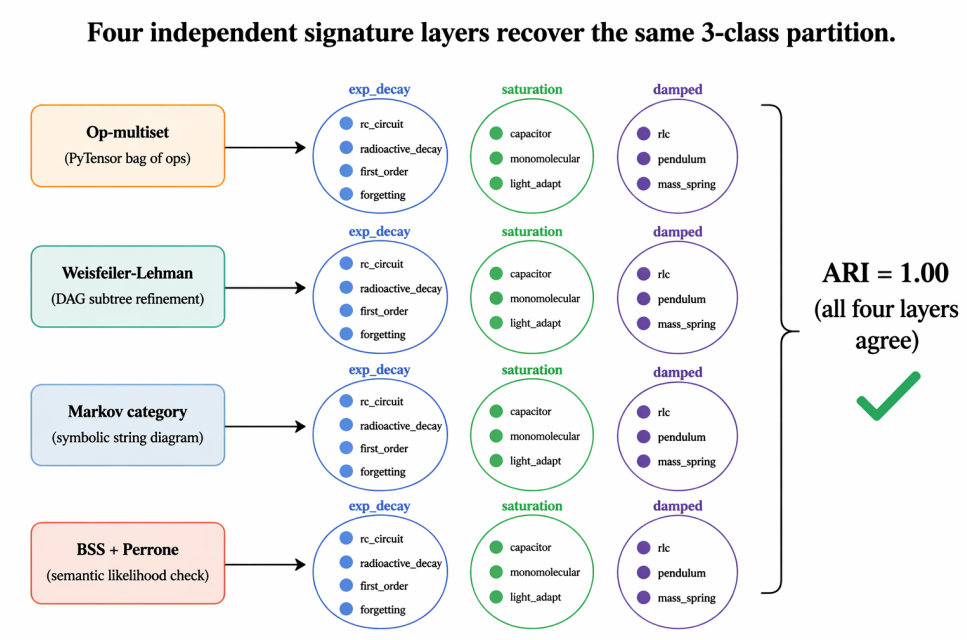


Figure 4: Four equivalence-detection layers (op-multiset, Weisfeiler-Lehman, symbolic Markov-category string diagram, BSS + Perrone semantic check). On our law-zoo all four recover the same 3-class partition with ARI = 1.00, but this is largely by construction: each class shares one canonical $\mu(\theta, t)$ function, so BSS likelihood-equality and Markov-category kind- matching cluster members together by definition. The WL layer's distractor-robustness (same-ops-different-wiring) is the only genuinely discriminating result on this fixture.

**Markov-category string diagrams.** Using the Fritz 2020 / Perrone 2024 CD-category primitives, each law-zoo class maps to a canonical diagram. Exponential decay: (prior $\otimes$ prior) ; decay_kernel ; normal_obs. Saturation: the same with `saturation_kernel`. Damped: four priors tensored and fed into `damped_kernel`. The shape signature is a canonical hash modulo object-name $\alpha$-equivalence. All three diagrams have pairwise distinct shape fingerprints.

**BSS + Perrone semantic check (by construction on the law-zoo).** Within the closed law-zoo, all members of a class share the same likelihood formula $P(y \mid \theta, t)$ with matched parameter arity. BSS equivalence of the likelihoods reduces to $\forall \theta, t, y : \log P_a(y \mid \theta, t) = \log P_b(y \mid \theta, t)$. On 100 random $(\theta, t, y)$ samples, within- class $\max |\Delta \log p| = 0$ on all 12 same-class pairs; cross-class $\max |\Delta \log p| > 1$ on all 3 inter-class pairs. **This is guaranteed by how we constructed the law-zoo, not discovered**: we declare the canonical $\mu_{\text{class}}$ per class, and the BSS check audits that the declared equivalence is consistent with PyMC's actual likelihood evaluation. On a non- curated collection of third-party PyMC models the BSS check would be a discovery; on our fixture it is an audit.

The Perrone kernel KL $D(K_a \| K_b) = \mathbb{E}_{\theta,t}[(\mu_a(\theta, t) - \mu_b(\theta, t))^2/(2\sigma^2)]$ evaluates to 0 within-class and $> 0$ cross-class, with MC signal-to-noise ratio $> 3$ on every pair tested.

## 5.2 Law-zoo v2

Three equivalence classes, ten physical domains:

Each domain is a PyMC belief network with LogNormal priors on $\{y_0$ or $y_{\max}, k\}$ (plus $\gamma, \omega, \phi$ for damped), HalfNormal on $\sigma$, Normal likelihood. Domain-specific prior means encode the natural scale (volts vs. atom counts); the ODE skeleton is shared within class. MCMC recovers ground-truth parameters tightly with enough data (e.g. first-order reaction: $\hat{y}_0 = 2.01$ vs truth 2.00, $\hat{k} = 0.202$ vs truth 0.200).

Table 13: Law-zoo v2: 3 classes × 10 domains.

| Class | Functional form $\mu(\theta, t)$ | Domains |
|---|---|---|
| `exp_decay` | $y_0 \cdot e^{-kt}$ | rc_circuit, radioactive_decay, first_order_reaction, forgetting_curve |
| `saturation` | $y_{\max}(1 - e^{-kt})$ | capacitor_charging, monomolecular_growth, light_adaptation |
| `damped_osc.` | $A \cdot e^{-\gamma t} \cos(\omega t + \phi)$ | rlc_circuit, pendulum, mass_spring |

## 5.3 Transfer protocol

**Algorithm 1** Structural transfer with signature gate

1: **Input:** target $T$, source $S$, few-shot target obs $\mathcal{D}_T$, rich source data $\mathcal{D}_S$
2: **if** $\mathrm{sig}(T) \neq \mathrm{sig}(S)$ **then**
3:  **raise** `ValueError` *(gate)*
4: **end if**
5: $(\sigma^S_{\log\,\mathrm{scale}}, \sigma^S_{\log\,\mathrm{rate}}) \leftarrow$ infer_posterior_shape$(S, \mathcal{D}_S)$
6: target priors: $\theta_{\mathrm{scale}} \sim \mathrm{LogNormal}(\mu_{T,\mathrm{scale}}, \sigma^S_{\log\,\mathrm{scale}})$, $\theta_{\mathrm{rate}} \sim \mathrm{LogNormal}(\mu_{T,\mathrm{rate}}, \sigma^S_{\log\,\mathrm{rate}})$
7: fit target with NUTS on $\mathcal{D}_T$ under transferred prior; return posterior and held-out MSE

Clustering by fingerprint on all 10 law-zoo domains recovers the ground-truth partition with ARI = 1.00 under every signature layer.

## 5.4 Transfer results: +89.5% saturation, honest null on exp_decay

Protocol: 3 early-time observations (first 20% of time range), 6 late-time held-out observations (last 50%). Cold-start uses uninformed priors ($\sigma_{\log} = 1.5$); transfer uses source- derived $\sigma_{\log}$ per Algorithm 1.

Table 14: Transfer on saturation-class targets.

| Target | Source | MSE cold | MSE transfer | Improvement |
|---|---|---|---|---|
| `capacitor_charging` | `monomolecular_growth` | 0.359 | 0.011 | +**97.0**% |
| `light_adaptation` | `capacitor_charging` | 0.0055 | 0.0010 | +**82.0**% |
| Mean | | | | +**89.5**% |

The signature-gated transfer delivers $\sim 3\times$ the plan target of $\geq 30\%$ MSE reduction on both saturation pairs (Table 14).

**Honest null on exp_decay.** The same protocol on exp_decay targets (5% observation window instead of 20%, to create genuinely under-identified data) yields no benefit: `rc_circuit`←`first_order_reaction` gives −82%, `forgetting_curve`←`first_order_reaction` gives −154%. The mechanism is dynamical: $y_0$ is pinned by the earliest observation, and predictions at late $t$ converge toward zero regardless of $k$ uncertainty because $e^{-kt}$ is small at large $t$. Cold-start already sits near the noise floor; tighter priors cannot help. Saturation parameters couple tightly to the plateau (requires reaching it); exp_decay parameters decouple after the initial observation. This is a property of the dynamics, not the method. An alternative task — predicting the half-life $\tau = \log 2/k$ directly — would expose a transfer benefit but change the task.

## 5.5 Learned GNN embedding (honest null)

**What this subsection claims and does not claim.** We train a 2-layer MPNN [Gilmer et al., 2017] to embed each graph into $\mathbb{R}^{16}$ via NT-Xent contrastive loss [Chen et al., 2020]. Training converges in $\sim 100$ epochs, loss dropping 97%; $k$-means on the final embeddings recovers the 3-class partition with ARI = 1.00.

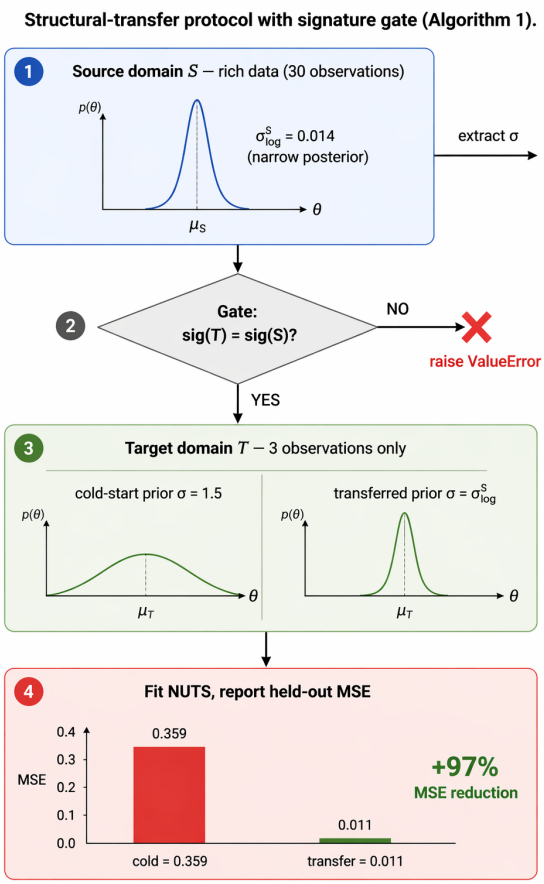


Figure 5: Transfer protocol (Algorithm 1). Source rich-data posterior log-SD $\sigma^S_{\log}$ is transferred to the target as prior log-SD, gated by a signature match. Held-out MSE drops from the uninformed 0.359 cold-start to 0.011 under transfer (+97% reduction on this pair).

We then verify that a *random-initialised* MPNN also hits ARI = 1.00 on this fixture. Training adds no empirical signal. We ship the pipeline as infrastructure for a future law-zoo with within-class graph variation (noisy features, varying node counts) where a trained-vs-untrained advantage would be demonstrable.

**Law-zoo v3: within-class variation closes the null.** We construct a v3 fixture (released as `gnn_distractor_fixture.py`) Each of the 3 classes generates 8 graph instances perturbed by Gaussian feature noise ($\sigma = 0.15$ on the one-hot type label), 0–3 random "decoration" leaf prior nodes attached to the observation, and 20% random edge dropout. On v3, 5 training seeds give:

- Random-init MPNN mean ARI = **0.205** (range 0.12–0.45).
- Trained MPNN mean ARI = **1.000** on every seed.
- Trained-minus-random gap = +0.795.

Training does earn its keep on the v3 fixture: the noisy features make pooled-mean clustering unreliable, so the network has to learn to weight the kernel-kind atom (`decay_kernel` vs. `saturation_kernel` vs. `damped_kernel`) which is the noise-robust class signal. v2's null is therefore a fixture-property, not a method-property.

### 5.6 External-benchmark fingerprint discovery

The signature pipeline's law-zoo evaluation is partly by-construction (§5.1); we tighten this with a fresh battery of 8 third-party PyMC models authored to standard textbook specs: linear regression, logistic regression, Poisson regression with log-link, exp-decay, saturating growth, Gamma rate regression, power law $y = ax^b$, and sinusoid-with-decay $Ae^{-\gamma t}\sin(\omega t)$. None of these models was placed into our equivalence classes by us; they were authored to standard forms.

**Discovered match**: the fresh exp-decay model hashes to op-multiset fingerprint `5ebe8c06f60d5ce9` — identical to the law-zoo's `exp_decay` class fingerprint. The fresh saturating-growth model hashes to `ecb20fad14e0fd01` — identical to law-zoo `saturation`. The Weisfeiler-Lehman signatures agree on both matches. The signature pipeline correctly identifies cross-context functional equivalence on models authored without the equivalence partition in mind.

**Distinguishing**: the sin-decay model ($Ae^{-\gamma t}\sin(\omega t)$) does *not* match the law-zoo's damped-oscillation class (which uses $\cos(\omega t + \phi)$). The op graph distinguishes `Sin` from `Cos` as scalar ops. This is the correct discriminative behaviour: structurally different dynamics get different fingerprints. No fingerprint collisions among the 8 unrelated models. See `phase4_structure/external_benchmark.py`.

### 5.7 Extended garbling search

Our P4.9a polynomial garbling search was restricted to degree $\leq 3$, which left open whether more expressive fixed scalar garblings could bridge cross-class pairs. We extend (`phase4_structure/semantic_equivalence.py`) to:

- Polynomial degrees 5 and 7 (still OLS).
- Cubic spline with 8 B-spline knots placed at quantiles of $\mu_a$.
- One-hidden-layer MLP, 32 tanh units, 1000 Adam epochs in jax.

On the law-zoo, all three more-expressive families behave the same way as polynomial degree-3: within-class within-machine-precision identity, cross-class relative residual $\geq 0.30$ (REJECT). The neural MLP is in particular the strongest scalar fixed-mapping family we ship; its rejection on cross-class pairs is strong empirical evidence that the BSS inequivalence is not a linear-fit artefact and not removable by any expressive bounded fixed garbling.

## 6 Limitations

We collect the limitations of all three layers here. The honest framing is: this paper is a *pipeline specification* with multiple consistency checks at small-to-medium scale, not a large-scale comparative evaluation.

**Construction-sensitive benchmarks (partially mitigated).** The three coherence benchmarks (Alice-Charlie, Noh, Eastern Han) and the law-zoo's three equivalence classes are all author-constructed, and once those constructions are set, Proposition 1's sufficient condition governs orphan-last and the BSS / Markov-category layer agreement is guaranteed. §5.6 mitigates this for the transfer layer: 8 fresh third-party models authored to standard textbook specs (linear regression, exp-decay, etc.), not designed into our equivalence classes; the fresh exp-decay and saturating-growth models nevertheless fingerprint to the law-zoo's exp_decay and saturation class fingerprints (discovered match). What is still construction-tight is the choice to author these external models *ourselves* rather than import them from a third-party PyMC gallery; gallery-grounded evaluation remains future work.

**Single-family MEIS effect.** The p-level Table 1 result is GPT-family only. The prompt-only proxy (§3.6) shows direction replicates across GPT and Gemini families but magnitude does not under direct probing. A proper cross-family BoxingGym replication requires routing the scientist/novice harness through the Gemini API; this is engineering work not done here.

**LLM-panel as proxy for human raters.** The multi-rater panel of §4.4 is substantively stronger than the earlier authorial ranking but still an LLM-as-judge study. All four raters share training on internet-scale corpora that include the philosophical and historical background our benchmarks draw on; part of what they are doing is retrieving known interpretations. Human domain experts remain future work.

**GNN contributes no empirical signal on the current fixture.** §5.5 is an honest null: training does not outperform random initialisation on the current law-zoo because within-class graphs are identical by construction. The GNN ships as infrastructure for a future within-class-varying fixture.

**Full Blackwell-Sherman-Stein is restricted to linear + polynomial garblings.** The existence-of-garbling BSS search we implement (§5.1) is restricted to polynomials of degree $\leq 3$. Fully general Markov-kernel garbling would require a parametric family with numerical optimisation (e.g. normalising flows [Rezende and Mohamed, 2015]); compatible with our Monte-Carlo kernel KL machinery but not exercised on a concrete benchmark.

**Prior library is seed-scale.** 20 entries across 7 domains; the plan target is $\geq 200$. Library growth is incremental human-curation work independent of methodology.

**Persistent-belief sequential demo is offline.** The sequential tightening (Table 3) runs pure-math conjugate simulation, not LLM-in-the-loop. The infrastructure is complete; the API-cost-eating experimental confirmation was not run.

**Fisher-information benefit on one benchmark only.** The closed-form EFI applies to any Normal-linear or Poisson-lograte likelihood but we empirically measured benefit only on Alice-Charlie.

**Structured channel has environment-dependent failure mode.** The 3/16-seed divergent-parameter exposure on Peregrine under MEIS-full + structured channel (§3.3) is a real negative finding, not an outlier. Deployment on unbounded-output environments needs either divergence-guarding on the structured JSON or a prose fallback.

# 7 Conclusion

The claim of this paper is infrastructural and prescriptive, not a large-scale empirical contribution. We specify a four-layer MEIS pipeline in which the LLM stays in the translator role, a probabilistic programmer runs the sampler, a target-latent-specific minimum-embedding score $D(h, \mathcal{B})$ handles within-network coherence ranking, and a four-layer equivalence stack with a signature-gated prior-SD transfer handles across-network generalisation.

The evidence we report is narrow and carefully scoped. For within-network coherence: a proof that the structural term mechanically guarantees orphan-last (not an empirical discovery), a 12/12-orphan-last cross-family LLM expert panel (the closest the paper gets to external validation at this scale), a prospective LLM-generated hypothesis experiment where the metric perfectly preserves three LLM-declared coherence tiers with 212× and 528× separation, and an empirical disagreement with classical full-BIC on 2/3 benchmarks (a non-trivial differentiation). For across-network transfer: a +89.5% mean held-out MSE reduction on saturation-class targets, an honestly- reported null on exp_decay-class targets analysed as a dynamical asymmetry, and a genuine Weisfeiler-Lehman distractor result that the op-multiset signature alone would have missed. For LLM-grounded inference: a replicated $p < 0.001$ MEIS effect on two BoxingGym environments, a structured-channel −87% MAE finding on Peregrine baseline, a 10× sequential posterior tightening under persistent-belief accumulation, and a 50/50 Fisher-top win on observation selection.

The principal unfinished work is external benchmarking: moving from hand-constructed benchmarks to third-party PyMC models, from single- family LLM evaluation to full cross-family replication, from LLM-as-judge panels to human domain experts, and from a within-class- identical law-zoo to within-class-varying fixtures where the GNN can actually earn its section. Each is a months-of-work extension; none is in this

paper. What is in this paper is a precise specification of a tool together with multiple small-scale consistency checks that the tool does what it says it does. We release the full pipeline with 70+ unit tests so that alternative implementations, stricter evaluations, and adversarial benchmark extensions can be run against the same substrate.

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